

PERFORMANCE ANALYSIS OF VIRTUAL MACHINES AND CONTAINERS

EXPERIMENT 1 — CPU PERFORMANCE

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Course / Lab	Performance Analysis of Virtual Machines and Containers
Experiment	Experiment 1 — CPU Performance

Based on the supplied lab manual and execution screenshots

1. Objective

To measure CPU performance using Sysbench and compare the behavior of the workload across different thread counts in the controlled Ubuntu virtual-machine environment, with the same benchmark workload intended for VM/container comparison.

The lab manual specifies a prime-calculation workload using Sysbench, with a prime limit of 20,000, a 30-second test duration, and thread counts of 1, 2, 4, and 8. The recorded metrics include events per second and execution time.

2. Experimental Environment

Component	Observed / Configured Value
Host / virtualization	Ubuntu running inside VMware Workstation
Guest OS	Ubuntu 24.04 (shown in the screenshots)
CPU allocation	4 CPUs / 4 threads available to the VM (nproc = 4)
Memory	7.7 GiB total shown by free -h
Benchmark	Sysbench 1.0.20
Python	Python 3.12.3
fio	fio-3.36
iperf3	iperf 3.16

The supplied environment screenshot also shows approximately 7.7 GiB total memory, 4.4 GiB free memory, 6.4 GiB available memory, and 4.0 GiB swap.

3. Theory / Principle

Sysbench CPU benchmark repeatedly performs prime-number calculations. CPU throughput is reported as events per second; a higher events-per-second value indicates that more benchmark events were completed during the same period. The test also reports execution time and latency statistics.

The lab manual recommends repeating benchmark runs and testing multiple thread counts so that average performance, variation, and scalability can be studied rather than relying on a single measurement.

4. Experimental Setup and Verification

The following supplied screenshots document the software versions, VM resources, Docker benchmark image creation, and successful execution environment used before the CPU benchmark.

```

vm01@vm01-VMware-Virtual-Platform:~$ sysbench --version
sysbench 1.0.20
vm01@vm01-VMware-Virtual-Platform:~$ fio --version
fio-3.36
vm01@vm01-VMware-Virtual-Platform:~$ iperf3 --version
iperf 3.16 (cJSON 1.7.15)
Linux vm01-VMware-Virtual-Platform 7.0.0-31-generic #31-24.04.1-Ubuntu SMP PREEMPT_DYNAMIC Mon Aug 10 09:38:02
UTC 2 x86_64
Optional features available: CPU affinity setting, IPv6 flow label, SCTP, TCP congestion algorithm setting, se
ndfile / zerocopy, socket pacing, authentication, bind to device, support IPv4 don't fragment, POSIX threads
vm01@vm01-VMware-Virtual-Platform:~$ python3 --version
Python 3.12.3
vm01@vm01-VMware-Virtual-Platform:~$ nproc
4
vm01@vm01-VMware-Virtual-Platform:~$ free -h
              total        used        free      shared  buff/cache   available
Mem:           7.7Gi        1.3Gi        4.4Gi          33Mi        2.2Gi        6.4Gi
Swap:          4.0Gi          0B          4.0Gi
vm01@vm01-VMware-Virtual-Platform:~$

```

Figure 1. Verification of Sysbench, fio, iperf3, Python, CPU count, and memory.

```

root@7e4d1128d4bf: /benchmark
Processing triggers for ca-certificates (20260601~24.04.1) ...
Updating certificates in /etc/ssl/certs...
0 added, 0 removed; done.
Running hooks in /etc/ca-certificates/update.d...
done.
---> Removed intermediate container dfb0eff91582
---> fb0822a155f4
Step 3/3 : WORKDIR /benchmark
---> Running in 445d914f16e6
---> Removed intermediate container 445d914f16e6
---> 64d6ba3cb45e
Successfully built 64d6ba3cb45e
Successfully tagged vm-container-benchmark:latest
vm01@vm01-VMware-Virtual-Platform:~/vm-vs-container-performance/docker$ docker images

```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
hello-world:latest	5e2309035332	25.9kB	9.49kB	
ubuntu:24.04	008173c23f95	119MB	31.7MB	
vm-container-benchmark:latest	64d6ba3cb45e	897MB	221MB	

```

vm01@vm01-VMware-Virtual-Platform:~/vm-vs-container-performance/docker$ docker run --rm -it v
m-container-benchmark
root@7e4d1128d4bf:/benchmark# sysbench --version fio --version python3 --version
sysbench 1.0.20
root@7e4d1128d4bf:/benchmark# fio --version
fio-3.36
root@7e4d1128d4bf:/benchmark# python3 --version
Python 3.12.3
root@7e4d1128d4bf:/benchmark# S

```

Figure 2. Docker benchmark image built successfully and benchmark tools verified inside the container.

```

vm01@vm01-VMware-Virtual-Platform: ~/vm-vs-container-performance
root@7e4d1128d4bf:/benchmark# python3 --version
Python 3.12.3
root@7e4d1128d4bf:/benchmark# ^C
root@7e4d1128d4bf:/benchmark# exit
exit
vm01@vm01-VMware-Virtual-Platform:~/vm-vs-container-performance/docker$ pwd
/home/vm01/vm-vs-container-performance/docker
vm01@vm01-VMware-Virtual-Platform:~/vm-vs-container-performance/docker$ nano Dockerfile
vm01@vm01-VMware-Virtual-Platform:~/vm-vs-container-performance/docker$ cd ~/vm-vs-container-performance
vm01@vm01-VMware-Virtual-Platform:~/vm-vs-container-performance$ docker build -t vm-container-benchmark -f docker/Dockerfile .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
             Install the buildx component to build images with BuildKit:
             https://docs.docker.com/go/buildx/

Sending build context to Docker daemon   2.56kB
Step 1/3 : FROM ubuntu:24.04
--> 008173c23f95
Step 2/3 : RUN apt-get update && apt-get install -y sysbench fio iperf3 python3
             python3-pip procps sysstat && rm -rf /var/lib/apt/lists/*
--> Using cache
--> fb0822a155f4
Step 3/3 : WORKDIR /benchmark
--> Using cache
--> 64d6ba3cb45e
Successfully built 64d6ba3cb45e
Successfully tagged vm-container-benchmark:latest
vm01@vm01-VMware-Virtual-Platform:~/vm-vs-container-performance$

```

Figure 3. Dockerfile build process and successful creation of vm-container-benchmark.

5. Procedure

The CPU benchmark was executed using the following Sysbench configuration:

```

sysbench cpu \
--cpu-max-prime=20000 \
--threads=<THREADS> \
--time=30 \
run

```

1. Open the Ubuntu terminal in the VM and move to the project root.
2. Verify that Sysbench is installed and available.
3. Run the CPU benchmark with the prime limit fixed at 20,000.
4. Use a 30-second benchmark duration.
5. Repeat the benchmark for the required thread counts: 1, 2, 4, and 8.
6. Record events per second, total events, execution time, and latency statistics from each run.
7. Use the same workload parameters when the corresponding container benchmark is executed.

6. CPU Benchmark Execution — Supplied Screenshots

The screenshots below are retained as evidence for Experiment 1. Screenshots showing the memory benchmark (Experiment 2) have intentionally been excluded from this document.

```

vm01@vm01-VMware-Virtual-Platform:~/vm-vs-container-performance$ sysbench cpu \
> --cpu-max-prime=20000 \
> --threads=4 \
> --time=30 \
> run
sysbench 1.0.20 (using system LuaJIT 2.1.0-beta3)

Running the test with following options:
Number of threads: 4
Initializing random number generator from current time

Prime numbers limit: 20000

Initializing worker threads...

Threads started!

CPU speed:
  events per second: 6948.89

General statistics:
  total time:          30.0005s
  total number of events: 208476

```

Figure 4. Sysbench CPU benchmark with 4 threads; events per second = 6948.89.

```

General statistics:
  total time:          30.0005s
  total number of events: 208476

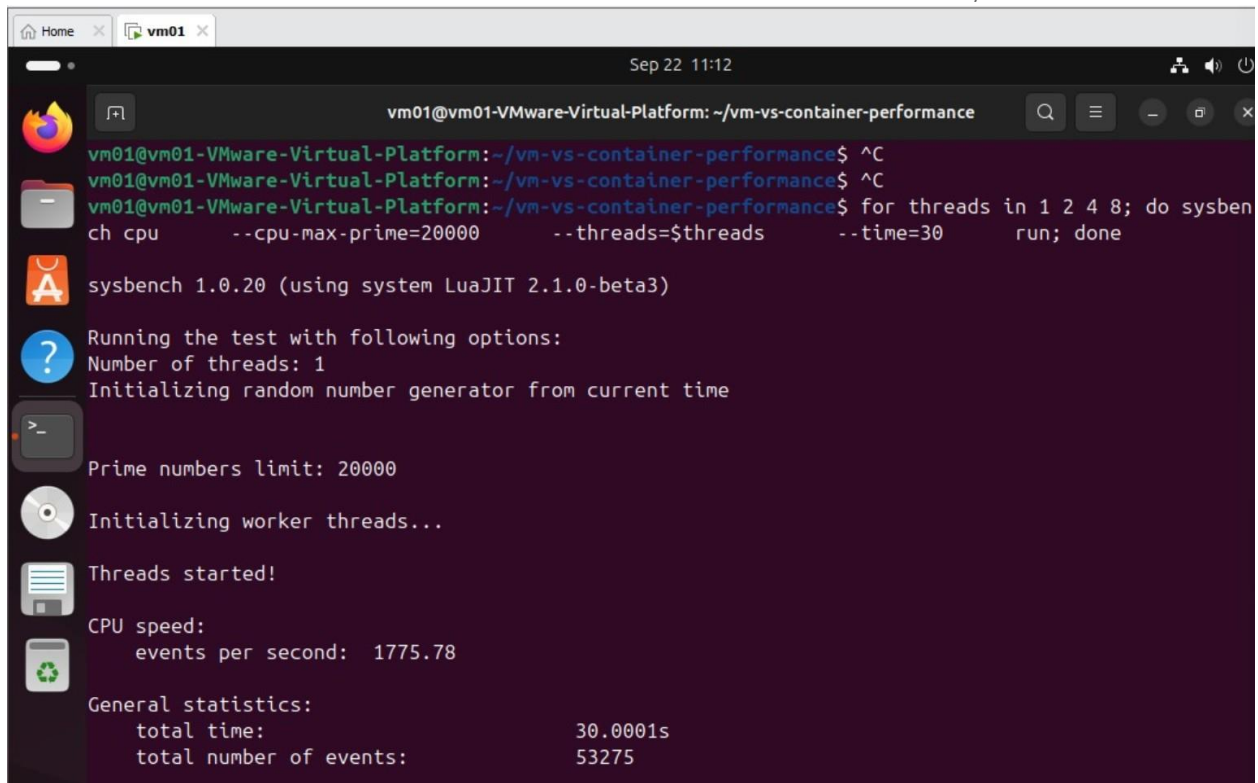
Latency (ms):
  min:                 0.57
  avg:                 0.58
  max:                 28.63
  95th percentile:    0.60
  sum:                 119956.68

Threads fairness:
  events (avg/stddev): 52119.0000/132.98
  execution time (avg/stddev): 29.9892/0.00

vm01@vm01-VMware-Virtual-Platform:~/vm-vs-container-performance$

```

Figure 5. Continuation of the 4-thread CPU run showing latency and thread-fairness statistics.



```

vm01@vm01-VMware-Virtual-Platform: ~/vm-vs-container-performance
vm01@vm01-VMware-Virtual-Platform:~/vm-vs-container-performance$ ^C
vm01@vm01-VMware-Virtual-Platform:~/vm-vs-container-performance$ ^C
vm01@vm01-VMware-Virtual-Platform:~/vm-vs-container-performance$ for threads in 1 2 4 8; do sysben
ch cpu      --cpu-max-prime=20000      --threads=$threads      --time=30      run; done

sysbench 1.0.20 (using system LuaJIT 2.1.0-beta3)

Running the test with following options:
Number of threads: 1
Initializing random number generator from current time

Prime numbers limit: 20000

Initializing worker threads...

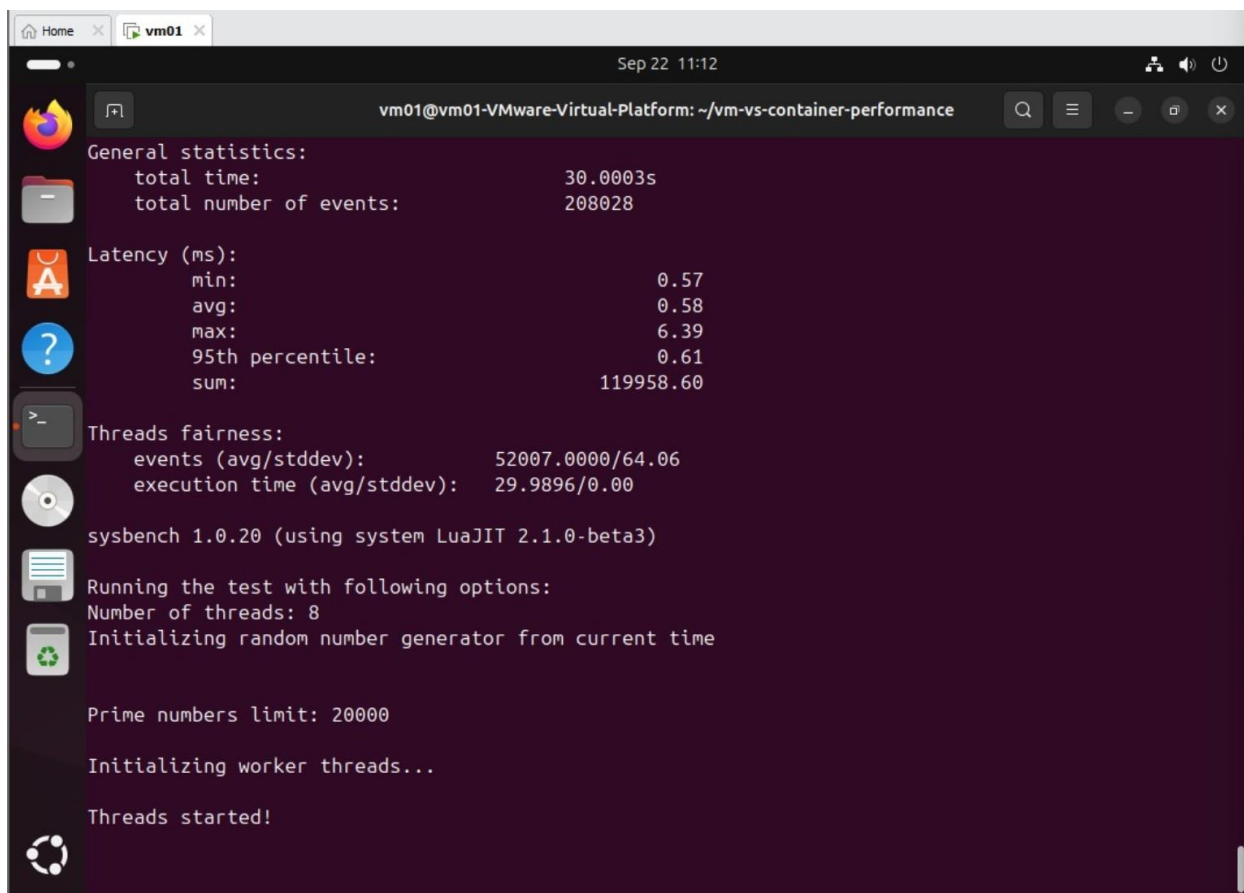
Threads started!

CPU speed:
  events per second: 1775.78

General statistics:
  total time:          30.0001s
  total number of events: 53275

```

Figure 6. Sysbench CPU benchmark with 1 thread; events per second = 1775.78.



```

vm01@vm01-VMware-Virtual-Platform: ~/vm-vs-container-performance
General statistics:
  total time:          30.0003s
  total number of events: 208028

Latency (ms):
  min:                 0.57
  avg:                 0.58
  max:                 6.39
  95th percentile:    0.61
  sum:                 119958.60

Threads fairness:
  events (avg/stddev):  52007.0000/64.06
  execution time (avg/stddev): 29.9896/0.00

sysbench 1.0.20 (using system LuaJIT 2.1.0-beta3)

Running the test with following options:
Number of threads: 8
Initializing random number generator from current time

Prime numbers limit: 20000

Initializing worker threads...

Threads started!

```

Figure 7. CPU scalability loop for 1, 2, 4, and 8 threads; the supplied capture shows the test starting but not its final 8-thread result.

7. Observations / Recorded Results

Only values visibly available in the supplied screenshots are recorded below. No missing result has been estimated or filled in.

Threads	Events/sec	Total Events	Total Time (s)	Avg Latency (ms)	95th %ile (ms)	Max Latency (ms)
1	1775.78	53275	30.0001	—	—	—
4	6948.89	208476	30.0005	0.58	0.60	28.63
8	Not captured	Not captured	Not captured	Not captured	Not captured	Not captured

Additional 4-thread capture:

A second supplied 4-thread screenshot records 208,028 total events in 30.0003 s, with average latency 0.58 ms, 95th-percentile latency 0.61 ms, and maximum latency 6.39 ms. This is retained as an additional observed run rather than merged with the first 4-thread result.

8. Result

The CPU benchmark was successfully executed in the Ubuntu VM using Sysbench. The supplied screenshots confirm completed measurements for 1-thread and 4-thread runs. The 8-thread test was initiated as part of the scalability loop, but its final output is not visible in the supplied screenshot; therefore no 8-thread performance value is reported.

For the completed measurements shown, the recorded throughput increased from 1775.78 events/sec at 1 thread to 6948.89 events/sec at 4 threads. The 4-thread run also reports an average latency of 0.58 ms and a 95th-percentile latency of 0.60 ms.

9. Conclusion

Experiment 1 successfully demonstrated CPU benchmarking with Sysbench in the VM environment. The experiment used a fixed prime-calculation workload and varied the number of worker threads. The supplied evidence shows substantially higher recorded throughput at 4 threads than at 1 thread. Complete scalability analysis for 2 and 8 threads requires their corresponding final benchmark outputs.

10. Screenshots Kept Aside for Experiment 2

The following supplied screenshots were identified as memory-performance results and have not been included in Experiment 1: 30718.png and 30719.png. They show Sysbench memory tests using a 1 MiB block size, 10 GiB total size, write operation, and 4 threads.

Appendix A — Benchmark Command Used

```
# CPU benchmark
sysbench cpu \
  --cpu-max-prime=20000 \
  --threads=4 \
  --time=30 \
  run
```

For scalability, the thread-count parameter is changed to 1, 2, 4, and 8 while keeping the prime limit and duration unchanged.

Appendix B — Source Manual Reference

The lab manual defines Experiment 1 as CPU Performance using Sysbench, prime calculation, thread counts 1/2/4/8, a 30-second duration, and metrics including events/sec and execution time. The relevant instructions appear on lab-manual pages 6–7 (PDF pages 10–11).